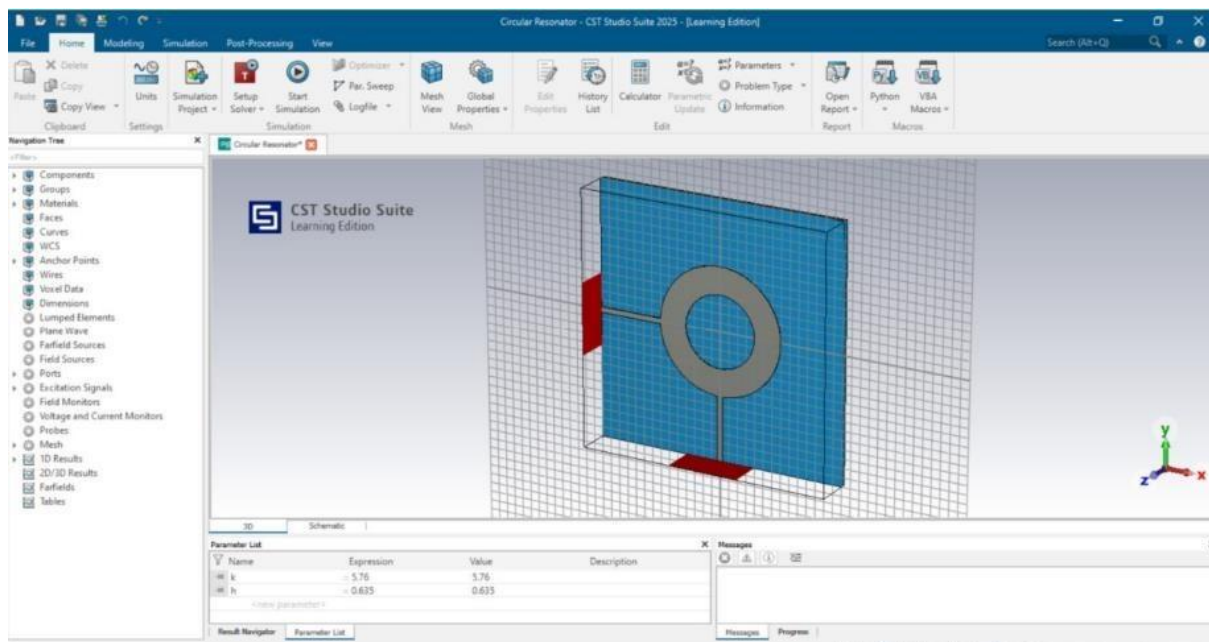


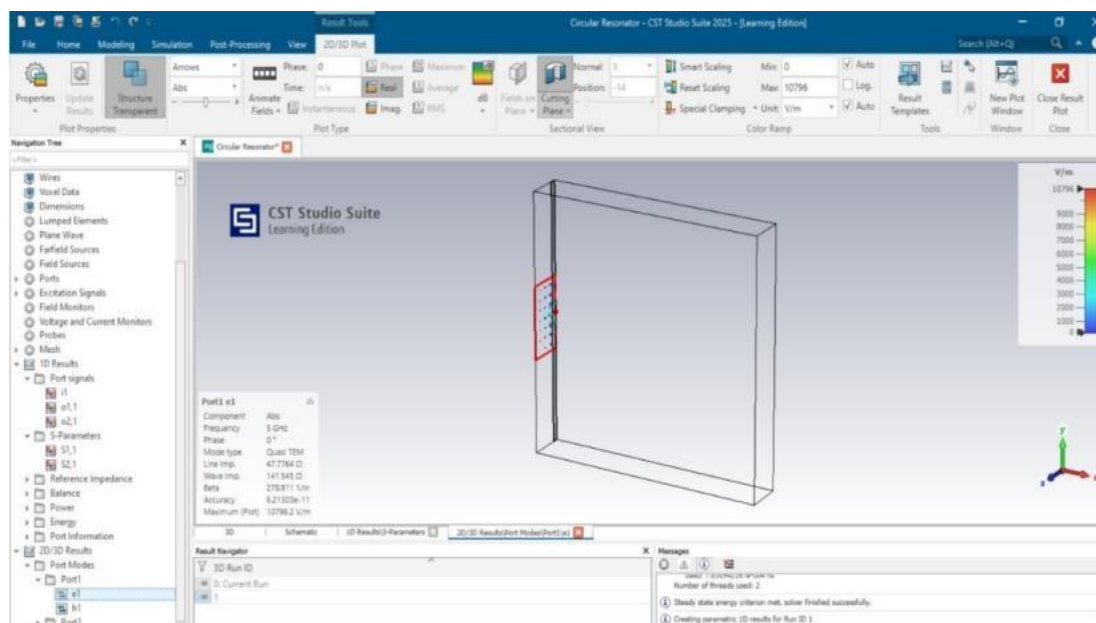
14. Electromagnetic Field and Power Distribution Analysis of a Circular Resonator in CST Studio Suite.

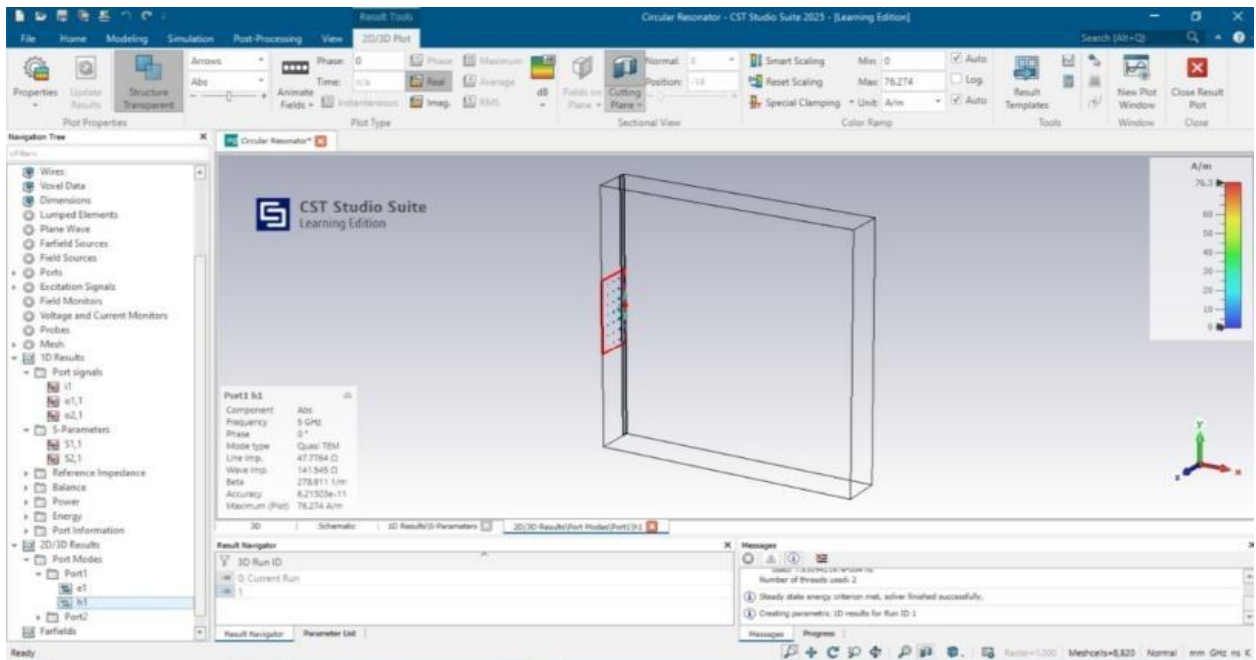
Test Case 1:

Analyse the electric field and magnetic field distribution at Port 1 of the circular resonator and examine the electric and magnetic field parameters at Port 1 for studying electromagnetic field behaviour and validating the system performance.



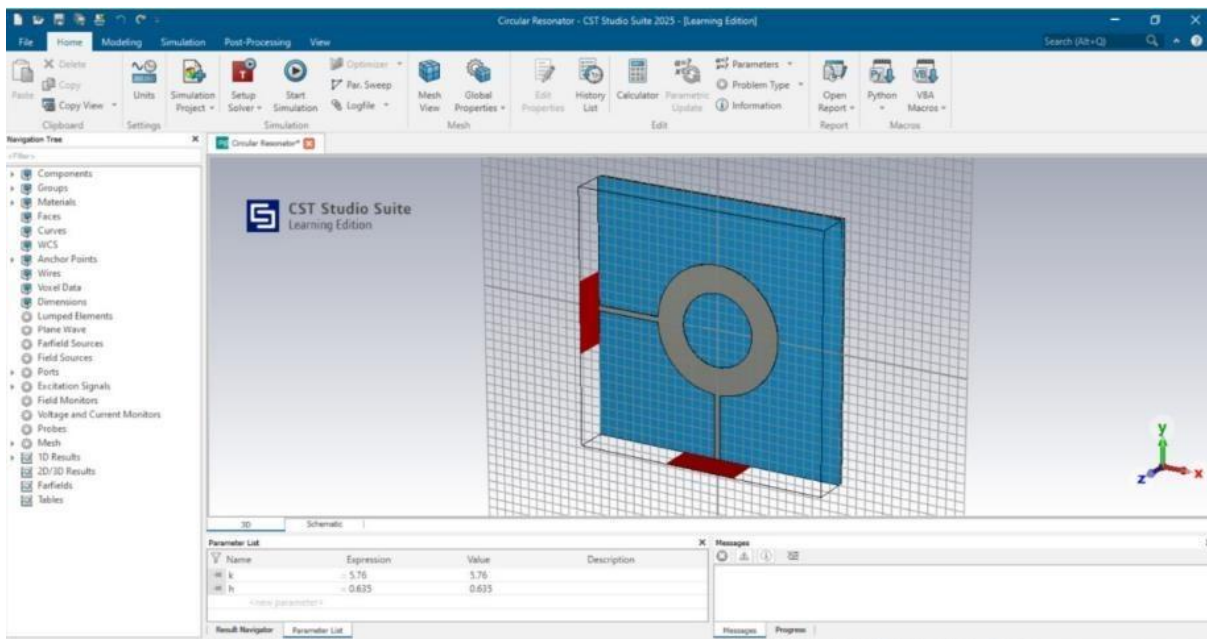
Output:



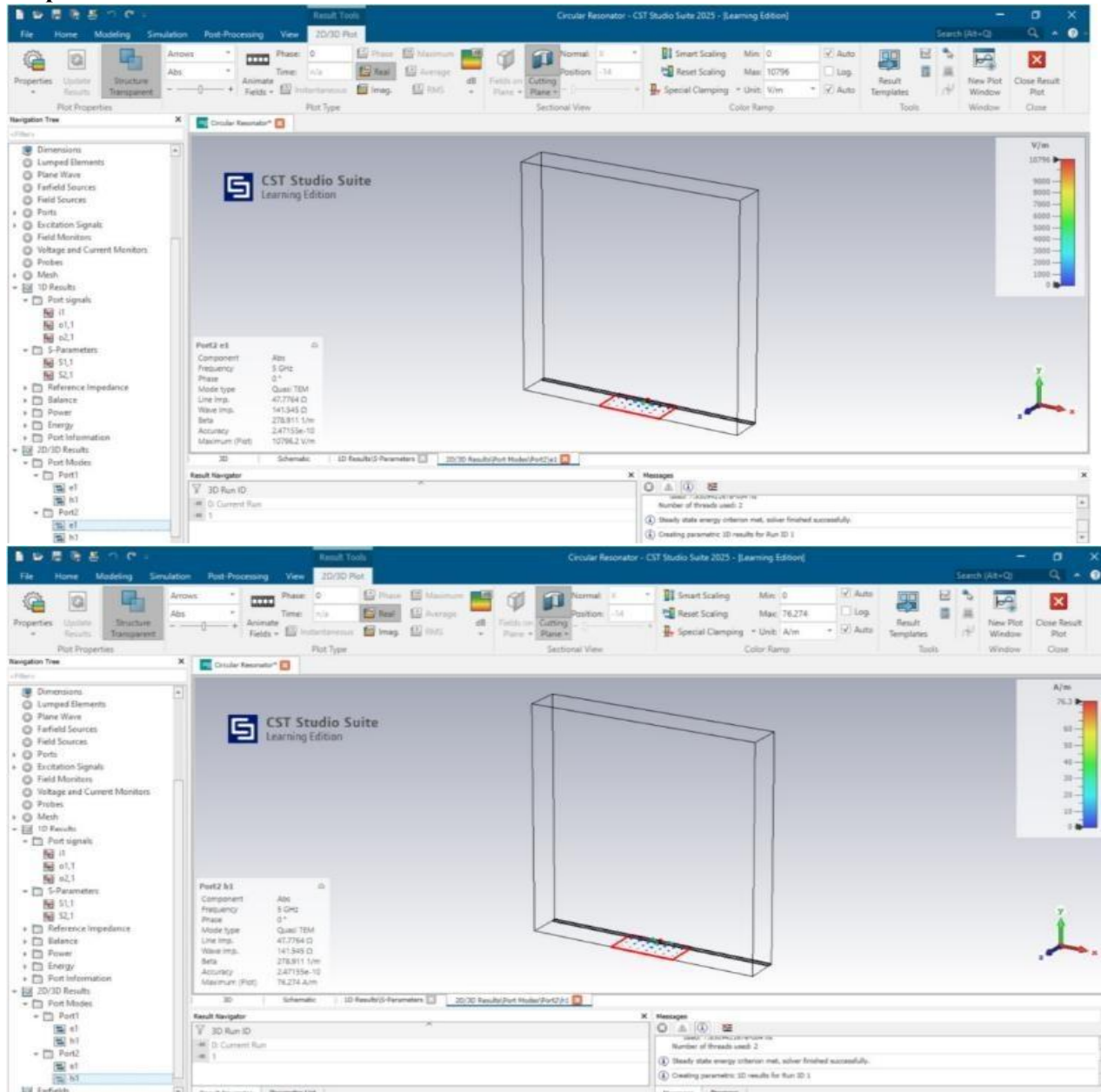


Test Case 2:

Analyze the electric field and magnetic field distribution at Port 2 of the circular resonator and examine the electric and magnetic field parameters at Port 2 for studying electromagnetic field behavior and validating the system performance.



Output:



RESULT:

The coaxial connector was successfully simulated using CST Studio Suite over the specified frequency range. The electromagnetic behavior of the structure was analyzed through power distribution among ports shows that the majority of the input power is transmitted to the output port with minimal reflection and negligible losses, indicating efficient signal transmission and satisfactory connector performance.